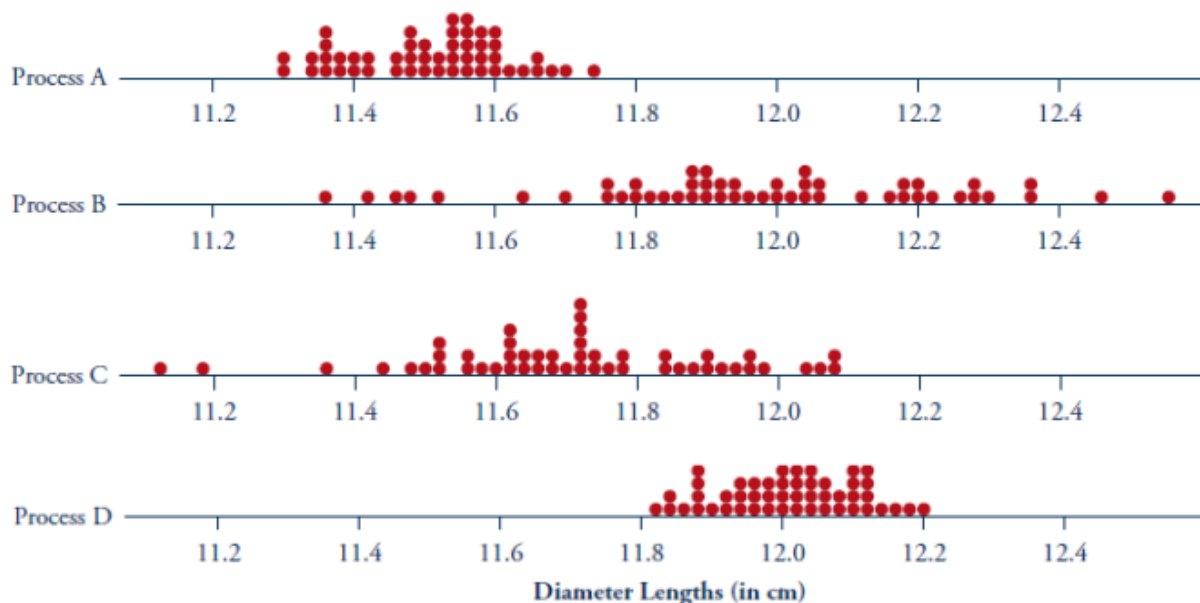


Answers: Comparing Processes: Rod Diameters

A process attempts to make rods with diameters of 12 cm, but the actual diameters vary slightly from rod to rod. Rods with diameters within 0.2 cm of the target value (between 11.8 and 12.2 cm) are considered to be within specifications (acceptable). Suppose that 50 rods are collected for inspection from each of four processes. Dotplots of their diameters are shown below.



a) Which process is best?

Process D. Its diameters are centered right around 12 cm, and almost all of them are between 11.8 and 12.2 cm.

b) Which process is most stable?

Process D. Its diameters run from about 11.83 to 12.21 cm, the narrowest spread. (Process A is also tightly clustered, but around the wrong value.)

c) Which process is least stable?

Process B. Its diameters are the most scattered, from about 11.36 to 12.55 cm, even though its center is near 12 cm.

d) Which process is generally farthest from the target?

Process A. Its diameters are centered around 11.55 cm, and none of them come within 0.2 cm of 12 cm.

Being on target is about the **center**. Being stable is about the **spread**. A process can be stable and still off target (A), or on target and unstable (B).